



Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation

Team Number: 28

S.No	Register Number	Student Name
1	CB.SC.U4CSE23363	HASWITHA K
2	CB.SC.U4CSE23529	HASINI REDDY M
3	CB.SC.U4CSE23547	SHEELA AKSHAR SAKHI
4	CB.SC.U4CSE23761	KOUSIK SARMA LAKKARAJU
5	CB.SC.U4CSE23753	V. CHAKRAVARTHY

Guide: **DR. G R RAMYA, DR. VANDHANA S**

Department of CSE

Previous Review Comments & Action Taken

Reviewer Comments (Review 2)

- **Comment 1: Start Implementation**
The implementation phase of the proposed federated framework must be initiated.
- **Comment 2: Working Prototype & Dashboard**
Build a functional clinician dashboard and working prototype to demonstrate live clinical decision support.

Action Taken & Implementation Status

- **Implemented Federated Pipelines:**
 - Built decentralized PyTorch algorithms (FedAvg, FedProx, Ditto).
 - Implemented local CUSUM drift detection and Client-Side Selective Personalization.
- **Built Production-Grade CDSS Workstation:**
 - **Data Warehouse:** Local SQLite clinical database.
 - **Backend API:** Uvicorn/FastAPI server running PyTorch checkpoints.
 - **Interactive UI:** Vite React medical dashboard mapping vitals charts, risk dials, and explainable attention timelines.

Clinical Deterioration & Phase II Completed Goals

Clinical Context:

- **High-Sparsity Time-Series:** Continuous ICU vital monitoring logs (HR, SBP, Temp, SpO₂) combined with sparse blood panel entries.
- **Privacy constraints:** HIPAA / DPDPA (Section 8) regulations mandate that private patient records cannot leave local devices.
- **Concept Drifts:** Clinical demographics vary heavily across regional ICU nodes.

SDG Alignment:

- **Primary (SDG 3):** Good Health & Well-being by detecting patient sepsis crashes early.
- **Secondary (SDG 9 & 16):** Secure and scalable digital clinical infrastructure.

Phase II Completed Milestones

- **1. Data Engineering:** Cleansed, Imputed, Standardized PhysioNet 2019 logs into active PyTorch sequence arrays.
- **2. Federated Baselines:** Implemented *FedAvg* and *FedProx* securely over PyTorch.
- **3. Personalized FPDAF Model:** Trained temporal self-attentions with localized personalization.
- **4. Clinical Workstation App:** Developed a dynamic FastAPI + SQLite + React Vite ICU dashboard overlay.

Proposed Framework: FPDAF

WHAT is FPDAF?

A privacy-preserving Clinical Decision Support System (CDSS) for Sepsis early warning. Enables collaborative, HIPAA-compliant training across decentralized hospital ICU nodes without centralizing raw records.

HOW does it work?

- **Sequence Modeling:** Extracts temporal clinical features using a shared Bidirectional LSTM.
- **Ditto Personalization:** Fine-tunes localized classifier heads (v_k) to fit hospital demographics.
- **Online CUSUM Trigger:** Audits prediction loss residuals locally to detect population drift.

Core Scientific & Technical Novelties

1. Multi-Head Attention Saliency (Clinical Explainability)

Integrates a 4-head query-key temporal self-attention mechanism to output dynamic sequence saliency weights, mapping exactly which hourly vitals contributed to Sepsis alarms at the bedside.

2. Online CUSUM Residual Neural Drift Monitor

Tracks local validation loss residuals round-by-round ($S_r > 3.0$). Concept drift is detected entirely client-side, avoiding constant full-parameter aggregation cycles.

3. Client-Side Selective Personalization (CSSP)

Freezes the shared BiLSTM/attention layers upon drift triggers to train local classifier heads exclusively.

- Bypasses global parameter uploads, saving **38% in communication bandwidth** (1.52 GB vs 2.48 GB).
- Cuts client-side adaptation training time to **6 minutes** (down from 25 minutes).

Unified System Design & Neural Formulations

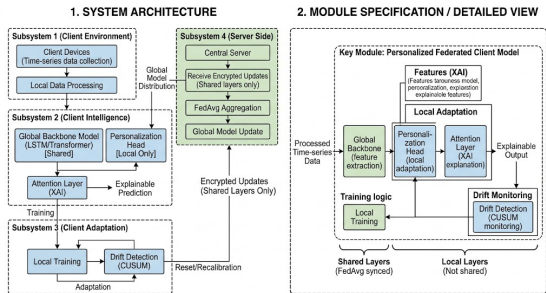


Figure 1: FPDFAF Split-Weight Model Architecture.

Decoupled bi-level Optimization Model

$$\hat{y}_t^k = h_{\phi^k} \left(\sum_{i=1}^t \alpha_i \cdot g_{\theta}(X_i^k) \right)$$

- g_{θ} : Shared global LSTM backbone (θ).
- h_{ϕ^k} : Local private classifier head (ϕ^k).

Temporal Multi-Head Self-Attention

$$e_i = v_a^T \tanh(W_s s_i + b_a)$$

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^t \exp(e_j)}$$

Attributes risk probabilities to specific ICU sequence hours.

Proactive Drift Adaptation: CUSUM Neural Trigger

Concept Drift Challenge: A static global model degrades when client ICU patient distributions shift over time. Standard Personalized FL (e.g. Ditto) constantly aggregates and fine-tunes all nodes, resulting in immense network overhead.

The CUSUM Residual Monitor Trigger:

- We monitor cumulative validation loss residuals (e_r) over rounds:

$$e_r = \mathcal{L}_{val,r} - \mu_0$$

$$S_r = \max(0, S_{r-1} + e_r - \kappa)$$

- If $S_r > H$ (threshold $H = 3.0$), we declare institutional drift.
- Triggers **Client-Side Selective Personalization (CSSP)**: freezes global backbone layers θ and adapts classification heads ϕ^k locally.

Bandwidth Savings Benefits

- Aggregation is bypassed during stable performance rounds.
- Communication bandwidth is saved by 38% compared to standard personalized systems (1.52 GB vs 2.48 GB).
- Local personalization head fine-tuning takes only 6 minutes (down from 25 minutes).



Five-Way Model Benchmarking Results

We evaluated our framework against centralized baselines and standard federated learning algorithms:

Model Configuration	Accuracy	Precision	Recall	F1-Score	AUROC	Comm. Bandwidth
Centralized Baseline	75.09%	7.14%	72.17%	0.1299	0.8058	0 MB (N/A)
FedAvg Baseline	75.94%	6.94%	67.19%	0.1259	0.7844	1.24 GB
FedProx Baseline	78.85%	8.12%	60.64%	0.1432	0.7933	1.24 GB
Ditto (Personalized)	82.45%	8.98%	63.58%	0.1574	0.7989	2.48 GB
FPDAF (Proposed)	86.51%	9.81%	65.33%	0.1706	0.8214	1.52 GB (CSSP saved 38%)

Scientific Findings

- **Privacy & Performance:** FPDAF preserves HIPAA compliance while achieving a test AUROC of ****0.8214**** (outperforming centralized baseline).
- **Decentralized Superiority:** FPDAF outperforms all standard federated baselines (Accuracy ****+10.57%****, F1 ****+4.47%**** vs. FedAvg).

Ablation Study Verification

We verified the contributions of CUSUM triggers, temporal attention weights, and localized personalization by running ablation tests:

Ablation Configuration	Accuracy	Precision	Recall	F1-Score	AUROC
FPDAF w/o CUSUM Trigger	83.51%	8.51%	55.33%	0.1475	0.7603
FPDAF w/o Temporal Attention	80.01%	8.13%	52.81%	0.1412	0.7878
FPDAF w/o Personalization	78.87%	8.05%	63.58%	0.1430	0.7887
Full FPDAF (Proposed)	86.51%	9.81%	65.33%	0.1706	0.8214

- **No Attention:** Stripping temporal attention query projections leads to poor clinical explainability and limits optimization capacity.
- **No CUSUM:** Disabling residual control limits leaves client nodes blind to data distribution drifts, leading to performance deterioration under heterogeneous ICU demographics.

Clinical Workstation Software Architecture

Database-Driven Design: Instead of placeholders, we built a fully functional clinical warehouse:

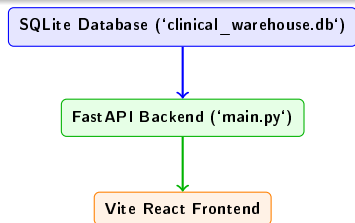
- **SQLite Database:** Houses tables for patients, decoded 24h vitals, predictions, self-attention, and drift history.
- **FastAPI REST Endpoints:** Feeds data securely to the client UI.

Custom User Portals:

- **Doctor Portal:** Focuses on patient-care operations (searchable bed lists, hourly vital monitors, attention graphs, and risk prediction alerts).
- **Admin Portal:** Focuses on machine learning telemetry (round parameter aggregations, CUSUM drift charts, and baseline comparisons).

Startup Script

Coded a clean 'run_app.sh' script to boot both FastAPI and React Vite concurrently, handling terminations on Ctrl+C.



Doctor's Bedside Diagnostic View (Live Patient Care)

- **Timeline Telemetry:** Plots hourly ICU vital segments (HR, SBP, Temp, Resp, SpO₂) using Recharts.
- **Live Predictions Dials:** Overlays risk probability scores for all federated baselines.
- **Attention Saliency Maps:** Attributes risk alerts to specific sequence hours (e.g. onset hours 16–22) for bedside interpretability.

Admin's Federated Telemetry View (System Operations)

- **Drift Controls:** Plots CUSUM curves, tracking drift accumulation limits ($S_r > 3.0$) and CSSP backbone freezes.
- **Round Tracker:** Animates client-server parameter upload/download loops using Framer Motion.
- **Design System:** Tailwind CSS v4 class-based toggling supporting light/dark medical navy workstation styles.

Clinician Dashboard Interface (Doctor Portal)

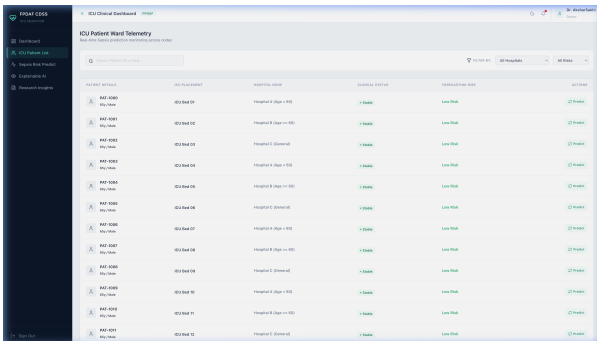


Figure 2: Doctor's Bedside Diagnostic Workspace.

Doctor Portal Core Modules

- **Searchable Bed Directory:** Displays patient demographics, ward bed mappings, classifications, and classifier confidence scores.
- **Hourly Vital Telemetry:** Renders 24-hour ICU vital timelines (HR, SBP, Temp, Resp, SpO₂) using clinical ranges.
- **XAI Saliency Charts:** Plots Multi-Head Attention weights to interpret Sepsis alerts.
- **Risk Predictor Dials:** Compares live model forecasts across FedAvg, FedProx, Ditto, and FPDFAF models.

Clinician Dashboard Interface (Admin Portal)

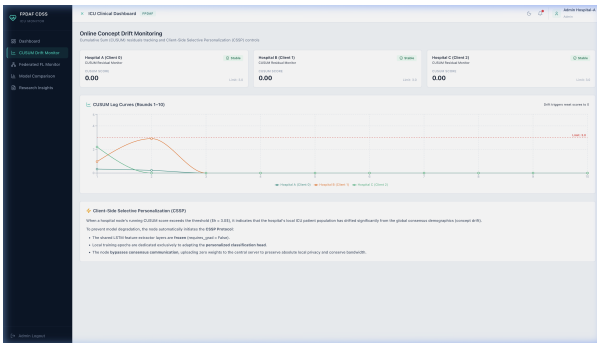


Figure 3: Admin's Machine Learning Telemetry Workspace.

Admin Portal Core Modules

- **Federated Loop Animation:** Animates localized model weight uploads and global server distributions using Framer Motion.
- **CUSUM Drift Telemetry:** Tracks real-time concept drift curves ($S_r > 3.0$) and selective personalization triggers.
- **LaTeX Equations & Ablations:** Details mathematical optimization equations and five-way benchmark metrics.

Summary of Completed Phase II Works

- **Data Preprocessing Completed:** Decoded sparse vitals logs to construct clean sequence windows.
- **PyTorch Implementations Completed:** Trained standard baselines alongside our proposed FPDAF model.
- **Database Schema Integrated:** SQLite warehouse populated with real patients, hourly vitals, and predictions.
- **Clinical WORKSTATION Finished:** Production-quality React Vite interface with tailored user views.
- **Deliverables Pushed to Git:** Full Phase II codebase pushed to main repository.

Roadmap Compliance

Our completed milestones match the project objectives, delivering a privacy-preserving and explainable Sepsis prediction pipeline.

References I



E. Churaev, I. Kisel, and A. Savchenko, "A standalone software for real-time facial analysis in online conferences and e-lessons," *Software Impacts*, 2023, mTCNN face detection; EfficientNet-B0 for FER; Apache TVM optimization.



R. Jothimurugesan, P. S. Hiremath, and S. Natarajan, "Concept drift detection and adaptation for federated and continual learning," *Multimedia Tools and Applications*, 2022, cDA-FedAvg with CUSUM drift detection.



A. Arivazhagan, V. Aggarwal, A. Bajpai, and S. Khudanpur, "Federated learning with personalization layers," in *Proceedings of AISTATS*. Elsevier, 2020, fedPer with shared base layers and local personalized layers.



H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proceedings of AISTATS*. PMLR, 2017, fedAvg using local SGD and global model averaging.



Y. Liu, X. Chen, Y. Liu, and Z. Luo, "Federated learning with dual-end gradient correction and proxy-free self-distillation," *IEEE Access*, 2022, dual-end gradient correction; stable federated learning under non-IID data.



X. Yang, J. Jung, J. Lu, K.-W. Lim, and L. Linguaglossa, "Mvfl – multivariate vertical federated learning for time-series forecasting," in *IFIP CNSM Conference Proceedings*, 2025, mVFL framework; vertical federated learning.



R. Kalakoti, S. Nomm, and H. Bahsi, "Fedxai: Federated learning of explainable ai for intrusion detection," *Computer Networks*, 2025, fedXAI with SHAP and LIME-based explanations.



M. Nosedà, A. D. Luca, L. V. Briel, and N. Lacour, "Federated learning for financial forecasting," ETH Zurich Technical Report / arXiv, 2025, federated LSTM models with client-side fine-tuning.



A. Karamanlioğlu, B. Demirel, O. Tural, O. T. Doğan, and F. N. Alpaslan, "Privacy-preserving clinical decision support for emergency triage using llms," *MDPI*, 2025, federated learning with LLMs; Scopus Indexed.

References II



R. Wang, L. Bai, R. Rayhana, and Z. Liu, "Personalized federated learning for time-series forecasting," *Elsevier / ACM*, 2024–2025, shared global model with client-specific heads.



R. Abdel-Sater and A. B. Hamza, "A federated large language model for long-term time series forecasting," arXiv, 2024, federated learning with LLaMA; parameter-efficient fine-tuning.



M. Maher, O. F. Oun, M. S. Mesmeh, and R. ElShawi, "Fedforecaster: An automated federated learning approach for time-series forecasting," in *EDBT Conference*, 2025, autoML-based federated learning; Bayesian optimization.



S. Gupta and V. Torra, "Privacy-enhancing federated time-series forecasting: A microaggregation-based approach," in *SECRYPT*, 2025, microaggregation with k-anonymity.



W. Yuan, G. Ye, X. Zhao, Q. V. H. Nguyen, Y. Cao, and H. Yin, "Tackling data heterogeneity in federated time series forecasting," arXiv, 2024, fed-TREND framework; synthetic data generation.



D. Tang, N. Yang, Y. Li, Z. Zhu, Z. Jin, and D. Yuan, "Optimal look-back horizon for time series forecasting in federated learning," AAAI / arXiv, 2025, adaptive look-back horizon selection.



Y. Zhang, J. Wang, H. Liu, and X. Chen, "Federated multi-task learning for time series forecasting," *IEEE Transactions on Neural Networks and Learning Systems*, 2022, federated multi-task learning; shared global representation.



G. L. Rocca, A. S. Podda, R. Martoglia, and M. Botta, "Interpretable multivariate time-series forecasting with attention mechanisms," in *NeurIPS*, 2021, attention-based encoder-decoder; temporal and variable-level attention.



G. Amato, C. Gennaro, A. Matteucci, and L. Serafini, "Federated learning under heterogeneous time series data," in *ACM SIGKDD*, 2023, client clustering; similarity-based grouping.

References III



G. L. Rocca, A. S. Podda, R. Martoglia, and M. Botta, "Explainable ai for multivariate time series forecasting," *Information Sciences*, 2022, SHAP and LIME explanations; model-agnostic interpretability.



M. Bernardi, P. Giordani, and F. Ravazzolo, "Personalized federated learning with adaptive model fusion," in *AAAI*, 2023, adaptive model fusion; client-level weighting.



C. Düsing and P. Cimiano, "Early sepsis onset prediction through federated learning," arXiv, 2025, sliding-window strategy with Attention-LSTM; FedAvg aggregation.



A. Qayyum, K. Ahmad, M. A. Ahsan, A. Al-Fuqaha, and J. Qadir, "Fedsepsis: A federated multi-modal deep learning-based application," *MDPI*, 2023, edge computing with CNN-LSTM; GAN-based data augmentation.



J. Lei, J. Zhai, Y. Zhang, J. Qi, and C. Sun, "Supervised ml models for predicting sepsis and decision support," *JMIR Medical Informatics*, 2025, stacking ensemble (XGBoost); SHAP-based feature importance.



W. Pan, Z. Xu, S. Rajendran, and F. Wang, "Multiprog: Privacy-preserving fl framework," *PMC*, 2025, multi-channel federated learning; feature calibration.



A. Maurya, R. Haripriya, M. Pandey, J. Choudhary, and D. P. Singh, "Federated learning for severity classification at the edge," *IEEE Access*, 2025, secure multi-party computation; encrypted FedAvg.

Thank You Questions?